Program overview

14-Dec-2022 11:24

Year	**2022/2023**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Minors Mechanical Engineering**

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Minors WB 2022			
WB-MI-168-22	**WB-MI-168-22 Robotica**		
Voor ET Bachelors			
IO3881-21	Design in Robotics	4	
TI3115TU	Software Engineering Methods	5	
WB3700	Robotics Minor Project	15	
WB3703	Introduction to ROS	2	
WB3705	Dynamics for Minor Robotics	4	
Voor IO Bachelors			
IO3881-21	Design in Robotics	4	
TI3105TU	Introduction to Python Programming	5	
WB2632	Advanced Engineering Design Project	6	
WB3700	Robotics Minor Project	15	
Voor WB Bachelors			
ET3033TU	Circuit Analysis	3	
IO3881-21	Design in Robotics	4	
TI3115TU	Software Engineering Methods	5	
WB3168	Robotic System Identification and Parameter Estimation	1	
WB3700	Robotics Minor Project	15	
WB3703	Introduction to ROS	2	
Voor TI Bachelors			
ET3033TU	Circuit Analysis	3	
IO3881-21	Design in Robotics	4	
WB3700	Robotics Minor Project	15	
WB3703	Introduction to ROS	2	
WB3704	Gazebo Assignment	2	
WB3705	Dynamics for Minor Robotics	4	
WB-MI-217-22	**WB-MI-217-22 Engineering for Large-Scale Energy Conversion and Storage**		
AE3516A	Fundamentals of Wind Energy I	3	
ET3034TU	Solar Energy	3	
WB3575	Energy Conversion: Devices, Systems and Efficiencies	4	
WB3580	Energy Storage	5	
WB3585	Assessment of Energy Systems	3	
WB3595	Design Project Renewables Based Energy Conversion and Storage	12	

Year	**2022/2023**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Minors Mechanical Engineering**

Minors WB 2022

Year	**2022/2023**
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Education	**Minors Mechanical Engineering**

WB-MI-168-22	
Contact for students	Martin Klomp
Program Title	Robotica
In association with the Faculty of	3mE
Program Goals	The student is able to develop a functional robot in a multidisciplinary team. This comprises the following technical competences: - executing a market analysis, designing the user experience (including external appearance), producing a slick robot (IO students already possess these competences) - calculating forces, torques, stiffness, strength, acceleration/deceleration, collisions and stability of the robot, and designing the mechanical parts and the drive train (WB students already possess these competences) - calculating currents, circuit dynamics, and heat production, selecting, reading, and interpreting sensor outputs, and developing, building and testing electronic circuits for power supply, safety, and data processing (EL students already possess these competences) - designing, implementing, and testing c++/python software modules for human interfacing, interaction, safety, and autonomous behavior of the robot (TI students already possess these competences). In addition, the students will learn how to implement the knowledge from their own BSc education in a large robotics project, and they learn to negotiate and discuss interdisciplinary trade-offs.
Intended for	Students from the following five TUD bachelor educations: - Mechanical Engineering - Industrial Design - Computer Science (Technische Informatica) - Electrical Engineering - Aerospace Engineering who want to build their own robot in a multidisciplinary team, and who want to learn some of the skills of their multidisciplinary teammates.
Gives access to	This is not a schakelminor. After following this minor, you will be well-prepared for robotics-related MSc research in your own discipline. For example, after this minor, the Mechanical Engineering BSc student is recommended to take an MSc in Mechanical Engineering and work on a robotics-related thesis project.
Expected prior Knowledge	Using Linux and programming in python. This will be taught during the minor, but prior experience is useful.
Prerequisites Minor	At least 90 ECTS in one of the four BSc educations mentioned above. Additionally, strong motivation is required; admission is limited to 10 students per discipline, due to limited availability of expensive high-tech equipment and intensive coaching.
Minor Coherence / Goal	The students follow 15 ECTS of carefully selected multidisciplinary courses (no courses in your own discipline), to get acquainted with the specialization of your team members.
Minor Content 1	Each participant follows 15 ECTS of courses, and completes 15 ECTS of teamwork projects. The course list depends on your BSc background so you will not follow everything. WB3701: Degisn in Robotics (IDE) WB3702: Design in Robotics (ME/AE/CSE/EE) ET3033TU: Circuit Analysis (ME/AE/CSE/IDE) TI3115TU: Software Engineering Methods (ME/AE/EE) TI3105TU: Introduction to Python (IDE) WB3703: Introduction to ROS (all) WB3704: Gazebo Assignment (CSE) WB3700: Robotics Minor Project (all) WB3705: Dynamics for Minor Robotics (CSE/EE) WB2632: Project Mechanica (IDE) WB3168: Robot Mechanics (ME/AE) In addition, there are various tutorials on design tools, software packages, and production methods, all aligned with the main projects listed below (teamwork projects for all students): Robotics Minor Project During the projects you will also actively participate other activities: - (Online) meetup with coaches - (Online) meetup with customers - Participate (online) in the final demo day
Education Methods	Lectures, tutorials and projects. A team assignment is a huge part of this minor. So working in teams and flexibility is needed.
Minor Assessment	Each part is assessed separately. The minor is successfully completed if the requirements of the bachelor's programme of the student are met.
Maximum number of participants	50
Minimum number of participants	20
Minor Remarks / Schedule	NOTE: Only 50 places are available. To be able to select students in order to create balanced teams, we have temporarily put the capacity at 70 students. After you enroll, we will ask you for a motivation letter by email. Thus, successful enrollment does not guarantee admission for this minor. Descriptions of the modules and the project can be found in the digital study guide.

Year	**2022/2023**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Minors Mechanical Engineering**

Voor ET Bachelors

IO3881-21	Design in Robotics	4
Course Coordinator	Dr.ing. M.C. Rozendaal	
Contact Hours / Week x/x/x/x		
Education Period	1 2	
Start Education	1	
Course Language	English	
Course Contents	In this IDE module, which is embedded within the Minor Robotics, we have bundled core IDE competencies within a coherent teaching program. The module teaches students how human and business aspects relate to and integrate with technology development. An iterative design methodology, inspired by SCRUM and the Basic Design Cycle (BDC), will be followed to foster this integration. This Module provides students with a parallel track (that will continue throughout the semester) to help them learn to apply human and business aspects in their project, across the different stages of technology development (i.e., from the fuzzy front-end to optimization).	
Study Goals	Students will learn how technology is used and experienced and by people and will learn how to position technology development in a business context. Further, students will be spurred on to adopt a critical perspective on technology. During the Minor these aspects are directly linked their prototyping activities that they will engage with throughout the Minor. Concerning methodology, students will learn about iterative design processes (based on SCRUM), and will be trained in how to communicate with others during this process and to collaborate in multidisciplinary setting that the Minor Robotics provides. Appropriate methods and tools from the Delft Design Guide (DDG) will be provided to students as a resource. We have developed 6 different tracks that will be explained in more detail below: 1. Human track: Understanding how technology is used and experienced by people. Contextual Inquiry, socio-technical systems, user-system interaction, co-design, user testing. 2. Business track: Understanding technology development in a business context. Value proposition, unique selling point, multistakeholder analysis, business modeling, entrepreneurship. 3. Prototyping track: Understanding prototyping as a vehicle for human, business technology integration. Experience prototyping, low fi-modeling (foam), Wizard of Oz approaches, interaction design/formgiving. 4. Critical thinking track: Understanding different viewpoints and values. Problem (re)framing, philosophy of technology, ethics. 5. Collaboration & Communication track: Understanding the importance of multidisciplinary teamwork. Teambuilding, presentation and communication, multidisciplinary collaboration, visualization. 6. Design Process track: Understanding iterative design processes. SCRUM methodology, Basic Design Cycle	
Education Method	This Module provides a parallel track in the Minor program. At different stages of their technology development, IDE knowledge and skills are provided that are appropriate and useful to them at that specific stage. The modules are organized in SCRUM sprints that are blocks of one or two weeks, in which a basic design cycle is followed. In each sprint a specific assignment is given to the students, that provides them with specific IDE knowledge and skills that they can directly apply.	
Literature and Study Materials	Delft Design Guide, van Boeijen et al. Integraal Product Ontwerpen, Roozenburg en Eekels	
Assessment	The means in which the students learning is assessed differs between courses. Assessment is done by reviewing the design outcomes/deliverables (prototypes, individual reports, personal reflections) both formative and summative.	
Special Information	Dr. ing. Marco Rozendaal, responsible teacher ID	

TI3115TU	Software Engineering Methods	5
Responsible Instructor	T. Durieux	
Contact Hours / Week x/x/x/x	4/0/0/0 lecture; 4/0/0/0 lab	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basics of (Python) programming or currently taking TI3105TU.	
Course Contents	In this course, students will learn: - software development processes, such as waterfall and agile - requirements engineering (use cases + user stories) - version control and Git - software testing - software design and design patterns - refactoring, code smells, and implementation best practices	
Study Goals	After this course, students will be able to: - Devise an entire life cycle of a software development process for their companies, choosing between different methodologies and ways of working (e.g., Scrum, Lean software development). - Write and evolve software requirement documents. - Perform automated software testing. - Understand the differences between a well-designed software and a not well-designed software. - Deliver useful working software.	
Education Method	Lectures + readings + software project.	
Literature and Study Materials	Slides and recorded videos.	
Assessment	90% project + 10% presentation, grades have to be higher than 5.0. We do not offer resits. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	This course is only accessible to students who have registered for the Computer Science minor programme (via the regular registration procedure).	
Co-Instructor	Prof.dr. A.E. Zaidman	

WB3700	Robotics Minor Project	15
Responsible Instructor	M. Klomp	
Contact Hours / Week x/x/x/x	x/x/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Building a robotic prototype for a real customer. This is the final project of the robotics minor. This is where all their learned objectives from the other courses in the minor should be visible and they should show what they have learned.	
Study Goals	After completing this course students should be able to: - work in a multidisciplinary team - contribute, from their own discipline, to a working robotic prototype	
Education Method	The students will be coached in working together in a multidisciplinairy team, contact with customers, technological problems.	
Assessment	In this course you will have multiple deliverables: a 'Detailed Design'-report (30%), a 'Handover document' (20%), the robotic prototype (40%) and a peer review of group members (10%).	
Department	3mE Department Cognitive Robotics	

WB3703	Introduction to ROS	2
Responsible Instructor	M. Klomp	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	An introduction to the Robot Operating System (ROS). This course consists of a couple of lectures (possibly online) followed by an assignment on a small mobile robot. > How to use ROS communication tools (topics, services, actions) to exchange information between functional modules > Visualization and creation of a custom environment with a robot > Mapping of the robot environment and navigation with a mobile robot > How to implement a pick-and-place function with industrial robot arms > Design of a complete robotic application with state machines	
Study Goals	After completing this course students should be able to: - list the core concepts of the ROS computation graph - use the ROS filesystem - install and use existing ROS packages - create a custom ROS package - control a mobile robot - control an industrial robot	
Education Method	Lectures (possibly online), followed by team assignment.	
Assessment	A final assignment will have to be handed in. This is a demo on a physical robot. The assignment will be done in pairs,	
Department	3mE Department Cognitive Robotics	

WB3705	Dynamics for Minor Robotics	4
Responsible Instructor	Dr.ing. L. Marchal Crespo	
Instructor	Ir. A.M. van der Niet	
Contact Hours / Week x/x/x/x	0.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Hibbeler Dynamics chapters; 12: kinematics of a particle 13: Force and acceleration (particle) 14: work and energy (particle) 15: impulse and momentum (particle) 16: planar kinematics (rigid body) 17: force and acceleration (rigid body, planar) 18: work and energy (rigid body, planar) 19: impulse and momentum (rigid body, planar) 22: vibrations	
Study Goals	1. Apply concepts and mathematical formulations related to kinematics of a point mass and of a rigid body (position, velocity, acceleration, in different reference frames with vector notation) 2. Formulate the equations of motion in different reference frames for point masses and rigid bodies, and solve these, using relevant mathematical formulations. 3. Formulate energy equations and solve these to describe the motion of point masses and rigid bodies. 4. Formulate linear and angular momentum equations to analyse situations where impulse and collisions play a role.	
Education Method	flipped classroom; watch lecture videos at home, instructional lectures at campus (not mandatory). Mandatory: 2 student presentations of selected exercises.	
Literature and Study Materials	Hibbeler Dynamics 14th edition SI units	
Assessment	written exam	
Department	3mE Department Cognitive Robotics	

Year	**2022/2023**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Minors Mechanical Engineering**

Voor IO Bachelors

IO3881-21	Design in Robotics	4
Course Coordinator	Dr.ing. M.C. Rozendaal	
Contact Hours / Week x/x/x/x		
Education Period	1 2	
Start Education	1	
Course Language	English	
Course Contents	In this IDE module, which is embedded within the Minor Robotics, we have bundled core IDE competencies within a coherent teaching program. The module teaches students how human and business aspects relate to and integrate with technology development. An iterative design methodology, inspired by SCRUM and the Basic Design Cycle (BDC), will be followed to foster this integration. This Module provides students with a parallel track (that will continue throughout the semester) to help them learn to apply human and business aspects in their project, across the different stages of technology development (i.e., from the fuzzy front-end to optimization).	
Study Goals	Students will learn how technology is used and experienced and by people and will learn how to position technology development in a business context. Further, students will be spurred on to adopt a critical perspective on technology. During the Minor these aspects are directly linked their prototyping activities that they will engage with throughout the Minor. Concerning methodology, students will learn about iterative design processes (based on SCRUM), and will be trained in how to communicate with others during this process and to collaborate in multidisciplinary setting that the Minor Robotics provides. Appropriate methods and tools from the Delft Design Guide (DDG) will be provided to students as a resource. We have developed 6 different tracks that will be explained in more detail below: 1. Human track: Understanding how technology is used and experienced by people. Contextual Inquiry, socio-technical systems, user-system interaction, co-design, user testing. 2. Business track: Understanding technology development in a business context. Value proposition, unique selling point, multistakeholder analysis, business modeling, entrepreneurship. 3. Prototyping track: Understanding prototyping as a vehicle for human, business technology integration. Experience prototyping, low fi-modeling (foam), Wizard of Oz approaches, interaction design/formgiving. 4. Critical thinking track: Understanding different viewpoints and values. Problem (re)framing, philosophy of technology, ethics. 5. Collaboration & Communication track: Understanding the importance of multidisciplinary teamwork. Teambuilding, presentation and communication, multidisciplinary collaboration, visualization. 6. Design Process track: Understanding iterative design processes. SCRUM methodology, Basic Design Cycle	
Education Method	This Module provides a parallel track in the Minor program. At different stages of their technology development, IDE knowledge and skills are provided that are appropriate and useful to them at that specific stage. The modules are organized in SCRUM sprints that are blocks of one or two weeks, in which a basic design cycle is followed. In each sprint a specific assignment is given to the students, that provides them with specific IDE knowledge and skills that they can directly apply.	
Literature and Study Materials	Delft Design Guide, van Boeijen et al. Integraal Product Ontwerpen, Roozenburg en Eekels	
Assessment	The means in which the students learning is assessed differs between courses. Assessment is done by reviewing the design outcomes/deliverables (prototypes, individual reports, personal reflections) both formative and summative.	
Special Information	Dr. ing. Marco Rozendaal, responsible teacher ID	

TI3105TU	Introduction to Python Programming	5
Responsible Instructor	F. Mulder	
Contact Hours / Week x/x/x/x	4/0/0/0 lecture; 4/0/0/0 lab	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	You are expected to have finished our Python Prerequisites mini-course before this course starts. Find it at https://ipp.pages.ewi.tudelft.nl/python-prerequisites/	
Course Contents	This course teaches programming in Python on an intermediate level.	
Study Goals	After completion of this course, the student is able to: 1. Implement Python programs based on a given natural-language description. 2. Write code that follows Python's code style guidelines. 3. Make use of basic data structures such as strings, lists, dictionaries, tuples and sets. 4. Apply object-oriented programming concepts such as inheritance, operator overloading and properties. 5. Apply functional programming concepts such as list comprehensions, recursion and higher-order functions, as well as iterators and generators. 6. Implement reading and writing data from/to different file formats. 7. Understand and write type hints.	
Education Method	Seminars and lab sessions.	
Computer Use	Students do all programming assignments on their own computers (except for the exams).	
Books	Think Python 2nd Edition by Allen B. Downey (freely available from thinkpython2.com)	
Assessment	Midterm exam (2 hours) in week 1.5, final exam (3 hours) in week 1.10. Both exams are computer exams. The midterm exam covers the course materials discussed until week 1.5, and the final exam covers all course materials. There is also a resit opportunity. The resit covers all course materials (there is no separate resit for the midterm exam). The final grade of the course is calculated as follows: course grade = 0.2*max(midterm, final) + 0.8*final So the midterm grade only counts if it is higher than the grade for the final exam. The final exam grade has to be at least 5.0. (And as usual, the course grade has to be at least 5.8 in order to pass the course.) The resit grade replaces the original grade only if the resit grade is greater than the original grade. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	You may use the Python documentation during the exam.	
Enrolment / Application	This course is only accessible to students who have registered for one of these minor programmes (via the regular registration procedure): Computer Science, Engineering with AI or Robotics.	
Co-Instructor	Prof.dr. A.E. Zaidman	

WB2632	Advanced Engineering Design Project	6
Responsible Instructor	Dr.ir. A. van Beek	
Instructor	Dr.ir. R.A.J. van Ostayen	
Contact Hours / Week x/x/x/x	8.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	Dutch	
Remarks	Hoorcollege online 4/0/0/0 Projecturen 4/0/0/0	
Department	3mE Department Precision & Microsystems Engineering	

WB3700	Robotics Minor Project	15
Responsible Instructor	M. Klomp	
Contact Hours / Week x/x/x/x	x/x/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Building a robotic prototype for a real customer. This is the final project of the robotics minor. This is where all their learned objectives from the other courses in the minor should be visible and they should show what they have learned.	
Study Goals	After completing this course students should be able to: - work in a multidisciplinary team - contribute, from their own discipline, to a working robotic prototype	
Education Method	The students will be coached in working together in a multidisciplinairy team, contact with customers, technological problems.	
Assessment	In this course you will have multiple deliverables: a 'Detailed Design'-report (30%), a 'Handover document' (20%), the robotic prototype (40%) and a peer review of group members (10%).	
Department	3mE Department Cognitive Robotics	

Year	**2022/2023**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Minors Mechanical Engineering**

Voor WB Bachelors

ET3033TU	Circuit Analysis	3
Responsible Instructor	B. Abdikivanani	
Instructor	B. Abdikivanani	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	The course requires familiarity with handling the complex numbers algebra.	
Course Contents	Electrical circuits fundamentals / basic concepts and components: current and charge, voltage and current sources, electrical circuits, circuit symbols, direct current, alternating current, resistors capacitors and inductors, Ohm's law, Kirchhoff's laws, power dissipation in resistors, series and parallel connections. Measurement of voltages and currents: common wave forms, measuring techniques and equipment. Resistance and DC Circuits: Thévenin and Norton theorems, superposition, nodal analysis, mesh analysis, (solving) circuit equations. Capacitance and electric fields: capacitors and capacitance, electric field strength and flux density, series and parallel connected capacitors, capacitor I/V relationship, sinusoidal voltage and current, energy stored in a capacitor. Inductance and Magnetic Fields: electromagnetism, reluctance, inductance, self-inductance, series and parallel connections, inductor I/V relationship, sinusoidal voltage and current, energy stored in an inductor, mutual inductance, transformers. Alternating voltages and currents: resistance vs. reactance, impedance, phasor diagrams, complex notation. Power in AC circuits: power dissipation in resistive circuits, power in capacitors and inductors, power in circuits with resistance and reactance, active and reactive power, power factor correction, three phase systems, power measurement. Frequency characteristics of AC circuits: two-port networks, the decibel, frequency response, filter networks, Bode diagrams, RLC circuits and resonance, stray capacitance and inductance. Transient behaviour: Charging of capacitors and energising of inductors, discharging and de-energising, first order systems, generalised response, second order systems, higher order systems.	
Study Goals	The course aims at building up insight and analysis instruments in the area of linear electrical circuits and components. It will also establish a solid foundation for understanding, analysing and synthesising electronic circuits using non-linear active components in both the analogue and digital domains.	
Education Method	The education is based on a self-study -> recap & Q/A -> class challenge for solving exercises -> discussion of the solutions cycle; there is a high emphasis on pre-preparation of online sessions and primarily do self-study.	
Literature and Study Materials	Neil Storey, "Electronics, a Systems Approach", 6th edition, Pearson Education, ISBN 978-0273773276. Material in the Brightspace learning environment: pitch handouts, links to online courses.	
Assessment	Homeworks + written exam	
Permitted Materials during Tests	The book electronic versions of the book are only allowed on laptops or e-readers, with the relevant device being set in flight-mode. Electronic copies on mobile phones are explicitly prohibited. In exceptional cases, the printout of the slides will be permitted. A standard electronic calculator is allowed, graphic calculators being only allowed in exceptional cases.	
Judgement	The final grade is calculated as a sliding-weighted average between the exam (Ex) and homeworks (Hw) grades; the final grade is rounded off to 0.5 points. A Hw=0 is accounted for in case no homework assignments are handed in. In case of a resit, the Hw grade obtained during the term remains valid.	
maximum aantal deelnemers	100	

IO3881-21	Design in Robotics	4
Course Coordinator	Dr.ing. M.C. Rozendaal	
Contact Hours / Week x/x/x/x		
Education Period	1 2	
Start Education	1	
Course Language	English	
Course Contents	In this IDE module, which is embedded within the Minor Robotics, we have bundled core IDE competencies within a coherent teaching program. The module teaches students how human and business aspects relate to and integrate with technology development. An iterative design methodology, inspired by SCRUM and the Basic Design Cycle (BDC), will be followed to foster this integration. This Module provides students with a parallel track (that will continue throughout the semester) to help them learn to apply human and business aspects in their project, across the different stages of technology development (i.e., from the fuzzy front-end to optimization).	
Study Goals	Students will learn how technology is used and experienced and by people and will learn how to position technology development in a business context. Further, students will be spurred on to adopt a critical perspective on technology. During the Minor these aspects are directly linked their prototyping activities that they will engage with throughout the Minor. Concerning methodology, students will learn about iterative design processes (based on SCRUM), and will be trained in how to communicate with others during this process and to collaborate in multidisciplinary setting that the Minor Robotics provides. Appropriate methods and tools from the Delft Design Guide (DDG) will be provided to students as a resource. We have developed 6 different tracks that will be explained in more detail below: 1. Human track: Understanding how technology is used and experienced by people. Contextual Inquiry, socio-technical systems, user-system interaction, co-design, user testing. 2. Business track: Understanding technology development in a business context. Value proposition, unique selling point, multistakeholder analysis, business modeling, entrepreneurship. 3. Prototyping track: Understanding prototyping as a vehicle for human, business technology integration. Experience prototyping, low fi-modeling (foam), Wizard of Oz approaches, interaction design/formgiving. 4. Critical thinking track: Understanding different viewpoints and values. Problem (re)framing, philosophy of technology, ethics. 5. Collaboration & Communication track: Understanding the importance of multidisciplinary teamwork. Teambuilding, presentation and communication, multidisciplinary collaboration, visualization. 6. Design Process track: Understanding iterative design processes. SCRUM methodology, Basic Design Cycle	
Education Method	This Module provides a parallel track in the Minor program. At different stages of their technology development, IDE knowledge and skills are provided that are appropriate and useful to them at that specific stage. The modules are organized in SCRUM sprints that are blocks of one or two weeks, in which a basic design cycle is followed. In each sprint a specific assignment is given to the students, that provides them with specific IDE knowledge and skills that they can directly apply.	
Literature and Study Materials	Delft Design Guide, van Boeijen et al. Integraal Product Ontwerpen, Roozenburg en Eekels	
Assessment	The means in which the students learning is assessed differs between courses. Assessment is done by reviewing the design outcomes/deliverables (prototypes, individual reports, personal reflections) both formative and summative.	
Special Information	Dr. ing. Marco Rozendaal, responsible teacher ID	

TI3115TU	Software Engineering Methods	5
Responsible Instructor	T. Durieux	
Contact Hours / Week x/x/x/x	4/0/0/0 lecture; 4/0/0/0 lab	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basics of (Python) programming or currently taking TI3105TU.	
Course Contents	In this course, students will learn: - software development processes, such as waterfall and agile - requirements engineering (use cases + user stories) - version control and Git - software testing - software design and design patterns - refactoring, code smells, and implementation best practices	
Study Goals	After this course, students will be able to: - Devise an entire life cycle of a software development process for their companies, choosing between different methodologies and ways of working (e.g., Scrum, Lean software development). - Write and evolve software requirement documents. - Perform automated software testing. - Understand the differences between a well-designed software and a not well-designed software. - Deliver useful working software.	
Education Method	Lectures + readings + software project.	
Literature and Study Materials	Slides and recorded videos.	
Assessment	90% project + 10% presentation, grades have to be higher than 5.0. We do not offer resits. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	This course is only accessible to students who have registered for the Computer Science minor programme (via the regular registration procedure).	
Co-Instructor	Prof.dr. A.E. Zaidman	

WB3168	Robotic System Identification and Parameter Estimation	1
Responsible Instructor	S. Dalla Gasperina	
Contact Hours / Week x/x/x/x	0/1/0/0	
Education Period	2	
Start Education	2	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Selected topics on sensors for robotics The course consists of 3 lectures 1. Sensors for Robotics (Navigation) a. Introduction to sensors (classification and characteristics) b. Visual sensors (2D Cameras) c. Depth sensors (Time-of-Flight) d. Proximity sensors (IR and ultrasonic) 2. Sensors for Robotics (Localization and Interaction) a. Beacon-based sensors (RFID and GPS) b. Position sensors (Encoders) c. Orientation sensors (Compass and IMU) d. Force/Torque sensors e. Pressure sensors f. Temperature sensors 3. Sensor for Robotics (Data Acquisition) a. Conditioning b. Amplification c. Filtering d. A/D Conversion	
Study Goals	The goal of this course is to provide an overview of sensors and their features for robotic applications.	
Department	3mE Department Cognitive Robotics	

WB3700	Robotics Minor Project	15
Responsible Instructor	M. Klomp	
Contact Hours / Week x/x/x/x	x/x/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Building a robotic prototype for a real customer. This is the final project of the robotics minor. This is where all their learned objectives from the other courses in the minor should be visible and they should show what they have learned.	
Study Goals	After completing this course students should be able to: - work in a multidisciplinary team - contribute, from their own discipline, to a working robotic prototype	
Education Method	The students will be coached in working together in a multidisciplinairy team, contact with customers, technological problems.	
Assessment	In this course you will have multiple deliverables: a 'Detailed Design'-report (30%), a 'Handover document' (20%), the robotic prototype (40%) and a peer review of group members (10%).	
Department	3mE Department Cognitive Robotics	

WB3703	Introduction to ROS	2
Responsible Instructor	M. Klomp	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	An introduction to the Robot Operating System (ROS). This course consists of a couple of lectures (possibly online) followed by an assignment on a small mobile robot. > How to use ROS communication tools (topics, services, actions) to exchange information between functional modules > Visualization and creation of a custom environment with a robot > Mapping of the robot environment and navigation with a mobile robot > How to implement a pick-and-place function with industrial robot arms > Design of a complete robotic application with state machines	
Study Goals	After completing this course students should be able to: - list the core concepts of the ROS computation graph - use the ROS filesystem - install and use existing ROS packages - create a custom ROS package - control a mobile robot - control an industrial robot	
Education Method	Lectures (possibly online), followed by team assignment.	
Assessment	A final assignment will have to be handed in. This is a demo on a physical robot. The assignment will be done in pairs,	
Department	3mE Department Cognitive Robotics	

Year	**2022/2023**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Minors Mechanical Engineering**

Voor TI Bachelors

ET3033TU	Circuit Analysis	3
Responsible Instructor	B. Abdikivanani	
Instructor	B. Abdikivanani	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	The course requires familiarity with handling the complex numbers algebra.	
Course Contents	Electrical circuits fundamentals / basic concepts and components: current and charge, voltage and current sources, electrical circuits, circuit symbols, direct current, alternating current, resistors capacitors and inductors, Ohm's law, Kirchhoff's laws, power dissipation in resistors, series and parallel connections. Measurement of voltages and currents: common wave forms, measuring techniques and equipment. Resistance and DC Circuits: Thévenin and Norton theorems, superposition, nodal analysis, mesh analysis, (solving) circuit equations. Capacitance and electric fields: capacitors and capacitance, electric field strength and flux density, series and parallel connected capacitors, capacitor I/V relationship, sinusoidal voltage and current, energy stored in a capacitor. Inductance and Magnetic Fields: electromagnetism, reluctance, inductance, self-inductance, series and parallel connections, inductor I/V relationship, sinusoidal voltage and current, energy stored in an inductor, mutual inductance, transformers. Alternating voltages and currents: resistance vs. reactance, impedance, phasor diagrams, complex notation. Power in AC circuits: power dissipation in resistive circuits, power in capacitors and inductors, power in circuits with resistance and reactance, active and reactive power, power factor correction, three phase systems, power measurement. Frequency characteristics of AC circuits: two-port networks, the decibel, frequency response, filter networks, Bode diagrams, RLC circuits and resonance, stray capacitance and inductance. Transient behaviour: Charging of capacitors and energising of inductors, discharging and de-energising, first order systems, generalised response, second order systems, higher order systems.	
Study Goals	The course aims at building up insight and analysis instruments in the area of linear electrical circuits and components. It will also establish a solid foundation for understanding, analysing and synthesising electronic circuits using non-linear active components in both the analogue and digital domains.	
Education Method	The education is based on a self-study -> recap & Q/A -> class challenge for solving exercises -> discussion of the solutions cycle; there is a high emphasis on pre-preparation of online sessions and primarily do self-study.	
Literature and Study Materials	Neil Storey, "Electronics, a Systems Approach", 6th edition, Pearson Education, ISBN 978-0273773276. Material in the Brightspace learning environment: pitch handouts, links to online courses.	
Assessment	Homeworks + written exam	
Permitted Materials during Tests	The book electronic versions of the book are only allowed on laptops or e-readers, with the relevant device being set in flight-mode. Electronic copies on mobile phones are explicitly prohibited. In exceptional cases, the printout of the slides will be permitted. A standard electronic calculator is allowed, graphic calculators being only allowed in exceptional cases.	
Judgement	The final grade is calculated as a sliding-weighted average between the exam (Ex) and homeworks (Hw) grades; the final grade is rounded off to 0.5 points. A Hw=0 is accounted for in case no homework assignments are handed in. In case of a resit, the Hw grade obtained during the term remains valid.	
maximum aantal deelnemers	100	

IO3881-21	Design in Robotics	4
Course Coordinator	Dr.ing. M.C. Rozendaal	
Contact Hours / Week x/x/x/x		
Education Period	1 2	
Start Education	1	
Course Language	English	
Course Contents	In this IDE module, which is embedded within the Minor Robotics, we have bundled core IDE competencies within a coherent teaching program. The module teaches students how human and business aspects relate to and integrate with technology development. An iterative design methodology, inspired by SCRUM and the Basic Design Cycle (BDC), will be followed to foster this integration. This Module provides students with a parallel track (that will continue throughout the semester) to help them learn to apply human and business aspects in their project, across the different stages of technology development (i.e., from the fuzzy front-end to optimization).	
Study Goals	Students will learn how technology is used and experienced and by people and will learn how to position technology development in a business context. Further, students will be spurred on to adopt a critical perspective on technology. During the Minor these aspects are directly linked their prototyping activities that they will engage with throughout the Minor. Concerning methodology, students will learn about iterative design processes (based on SCRUM), and will be trained in how to communicate with others during this process and to collaborate in multidisciplinary setting that the Minor Robotics provides. Appropriate methods and tools from the Delft Design Guide (DDG) will be provided to students as a resource. We have developed 6 different tracks that will be explained in more detail below: 1. Human track: Understanding how technology is used and experienced by people. Contextual Inquiry, socio-technical systems, user-system interaction, co-design, user testing. 2. Business track: Understanding technology development in a business context. Value proposition, unique selling point, multistakeholder analysis, business modeling, entrepreneurship. 3. Prototyping track: Understanding prototyping as a vehicle for human, business technology integration. Experience prototyping, low fi-modeling (foam), Wizard of Oz approaches, interaction design/formgiving. 4. Critical thinking track: Understanding different viewpoints and values. Problem (re)framing, philosophy of technology, ethics. 5. Collaboration & Communication track: Understanding the importance of multidisciplinary teamwork. Teambuilding, presentation and communication, multidisciplinary collaboration, visualization. 6. Design Process track: Understanding iterative design processes. SCRUM methodology, Basic Design Cycle	
Education Method	This Module provides a parallel track in the Minor program. At different stages of their technology development, IDE knowledge and skills are provided that are appropriate and useful to them at that specific stage. The modules are organized in SCRUM sprints that are blocks of one or two weeks, in which a basic design cycle is followed. In each sprint a specific assignment is given to the students, that provides them with specific IDE knowledge and skills that they can directly apply.	
Literature and Study Materials	Delft Design Guide, van Boeijen et al. Integraal Product Ontwerpen, Roozenburg en Eekels	
Assessment	The means in which the students learning is assessed differs between courses. Assessment is done by reviewing the design outcomes/deliverables (prototypes, individual reports, personal reflections) both formative and summative.	
Special Information	Dr. ing. Marco Rozendaal, responsible teacher ID	

WB3700	Robotics Minor Project	15
Responsible Instructor	M. Klomp	
Contact Hours / Week x/x/x/x	x/x/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Building a robotic prototype for a real customer. This is the final project of the robotics minor. This is where all their learned objectives from the other courses in the minor should be visible and they should show what they have learned.	
Study Goals	After completing this course students should be able to: - work in a multidisciplinary team - contribute, from their own discipline, to a working robotic prototype	
Education Method	The students will be coached in working together in a multidisciplinairy team, contact with customers, technological problems.	
Assessment	In this course you will have multiple deliverables: a 'Detailed Design'-report (30%), a 'Handover document' (20%), the robotic prototype (40%) and a peer review of group members (10%).	
Department	3mE Department Cognitive Robotics	

WB3703	Introduction to ROS	2
Responsible Instructor	M. Klomp	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	An introduction to the Robot Operating System (ROS). This course consists of a couple of lectures (possibly online) followed by an assignment on a small mobile robot. > How to use ROS communication tools (topics, services, actions) to exchange information between functional modules > Visualization and creation of a custom environment with a robot > Mapping of the robot environment and navigation with a mobile robot > How to implement a pick-and-place function with industrial robot arms > Design of a complete robotic application with state machines	
Study Goals	After completing this course students should be able to: - list the core concepts of the ROS computation graph - use the ROS filesystem - install and use existing ROS packages - create a custom ROS package - control a mobile robot - control an industrial robot	
Education Method	Lectures (possibly online), followed by team assignment.	
Assessment	A final assignment will have to be handed in. This is a demo on a physical robot. The assignment will be done in pairs,	
Department	3mE Department Cognitive Robotics	

WB3704	Gazebo Assignment	2
Responsible Instructor	M. Klomp	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	In this course the student will learn about the use of the Gazebo ssimulator in ROS. Firstly creating your own model of the robot, and then controlling the robot, reading out sensor values, and interacting with the environment.	
Study Goals	After this course the student will be able to: - Understand the importance of a simulation of the robot. - Build a working simulation of an actual robot inclusing sensors and actuators.	
Education Method	One 45 min lecture, followed by an assignment.	
Assessment	The final assignment of this course will be a working Gazebo simulation of a to be designed robot.	
Department	3mE Department Cognitive Robotics	

WB3705	Dynamics for Minor Robotics	4
Responsible Instructor	Dr.ing. L. Marchal Crespo	
Instructor	Ir. A.M. van der Niet	
Contact Hours / Week x/x/x/x	0.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Hibbeler Dynamics chapters; 12: kinematics of a particle 13: Force and acceleration (particle) 14: work and energy (particle) 15: impulse and momentum (particle) 16: planar kinematics (rigid body) 17: force and acceleration (rigid body, planar) 18: work and energy (rigid body, planar) 19: impulse and momentum (rigid body, planar) 22: vibrations	
Study Goals	1. Apply concepts and mathematical formulations related to kinematics of a point mass and of a rigid body (position, velocity, acceleration, in different reference frames with vector notation) 2. Formulate the equations of motion in different reference frames for point masses and rigid bodies, and solve these, using relevant mathematical formulations. 3. Formulate energy equations and solve these to describe the motion of point masses and rigid bodies. 4. Formulate linear and angular momentum equations to analyse situations where impulse and collisions play a role.	
Education Method	flipped classroom; watch lecture videos at home, instructional lectures at campus (not mandatory). Mandatory: 2 student presentations of selected exercises.	
Literature and Study Materials	Hibbeler Dynamics 14th edition SI units	
Assessment	written exam	
Department	3mE Department Cognitive Robotics	

Year	**2022/2023**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Minors Mechanical Engineering**

WB-MI-217-22	
Contact for students	Prof.dr.ir. Wiebren de Jong Section Large-Scale Energy storage (LSE) Faculty 3mE - Department of Process & Energy Leeghwaterstraag 39, office 34 J-0-920 (lab of Process & Energy) Telephone: 015-2789476 Mobile phone: 06-51236425 Mail: wiebren.dejong@tudelft.nl
Program Title	ELECS
Director of Education	Dr. R. Delfos Faculty 3mE r.delfos@tudelft.nl
EC Program	30 ECTS
Program Start (Study Year)	2019
Introduction 1	Energy conversion is the engine of the economics of our society. For a long time since the industrial revolution in the 18th century fossil fuels have dominated energy supply to a multitude of industries, the transportation sector and households. Reduction of CO2 emissions is urgently needed in view of Climate Change. Therefore, alternative energy supply is coming up rapidly. More and more, electricity will form the energy carrier for processes and household heat and power supply. Solar and wind power supply are increasingly implemented and become cheaper in fast pace. However, solar and wind are intermittent in nature and their power supply characteristics do not match the demand, neither in time nor place. There are periods and places with excess electricity supply compared to the demand, but certainly also times and locations with power supply shortage. Energy storage therefore is needed. In order to seriously reduce our societies' dependency on fossil fuels a huge effort is currently made to develop, scale-up and implement energy storage systems. Such systems are required for both short term (daily fluctuations and day-night rhythm) and long term (bridging reliable energy supply over seasons). This minor 'ELECS' covers both the aspects of renewable energy supply, energy conversion and storage and system integration
Program Goals	The aim of the minor is to equip bachelor engineering students with selected knowledge and skills concerning energy conversion and storage systems and component technologies that are based on renewable resources, which are as such mostly fluctuating and not-well predictable. Electricity is not easily and cost-effectively stored, so a mismatch both in time and geographically- between demand and supply of power is a real societal challenge. Also, storage of electricity needs to be accomplished at widely varying time scales to cope with fast weather fluctuations, day-night rhythm, occurrence of larger weather systems effects and seasonal variation of supply. This minor will deal with providing the fundamentals of a the most relevant intermittent renewable energy supply systems, wind and solar energy systems. Furthermore, systems analysis performance evaluation courses (energy storage, life cycle assessment) and practical studies (system modelling of energy conversion and storage systems and component level study design/research project) will be offered to enable students to make an evaluation of a large renewables based energy conversion and storage system. The difference with other existing minors that are offered within TU Delft is that a group of renewable energy systems is considered here in the framework of the challenge of storing energy at different time scales, taking a systems evaluation and practical component study approach together, which is unique. In summary, the topic of the minor is highly relevant due to the energy transition the world faces. Because the minor provides a comprehensive introduction to renewable energies, providing fundamental and practical knowledge and skills to students in this domain, it will be attractive and challenging to our engineering students.
Administration by the Education of	Faculty 3mE Mechanical Engineering
Minor Title	Engineering for Large-scale energy conversion and storage ELECS
EC Minor (Volume)	30 ECTS
Intended for	Targeted students for the Minor ELECS are those with an engineering basic knowledge of thermodynamics, heat transfer and/or fluid mechanics; i.e. students of Mechanical Engineering, Aerospace Engineering, Applied Sciences, Electrical Engineering, Technology Policy and Management, and Civil Engineering (Strictly: MT, Wb, CT, EE, TA, AE, TB, LST, MST, NB, TN)
Minor Content 1	The minor ELECS will provide students with an attractive constellation of curriculum courses and a multi-disciplinary integrated project concerning the topic of energy conversion and storage engineering. Aspects of variability in power (and heat) production characteristics and demand matching issues will be dealt with, combined with the evaluation of conversion characteristics and energy system evaluation. The undergraduate student having followed this Minor will be able to: - Understand and apply knowledge of renewable power and heat supply systems - Apply process calculations for energy conversion and storage systems based on thermodynamics, mass and heat transfer and fluid mechanics - Evaluate systems based on the concepts of exergy analysis and life cycle assessment - Provide insights into current state of the art and future trends in energy engineering - Synthesize their knowledge and competences in a final design project, including aspects of technical communication (presentations, technical report writing). Targeted students for the Minor ELECS are those with an engineering basic knowledge of thermodynamics, heat transfer and/or fluid mechanics; i.e. students of Mechanical Engineering, Aerospace Engineering, Applied Sciences, Electrical Engineering, Technology Policy and Management, and Civil Engineering (Strictly: MT, Wb, CT, EE, TA, AE, TB, LST, MST, NB, TN)
Remarks Maximum number of participants	75

AE3516A	Fundamentals of Wind Energy I	3
Responsible Instructor	Dr.ir. A.C. Viré	
Instructor	Dr. D.A. von Terzi	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course introduces the topic of offshore wind energy by going through the main steps that a wind farm developer needs to achieve in order to build an offshore wind farm. The main topics cover in the course are: - Historical developments in wind energy - Wind energy harvesting and Betz limit - Offshore wind and wave conditions - Estimation of the energy yield - Design considerations for wind turbine farms - Economic aspects - Planning issues - Emerging concepts such as floating wind turbines	
Study Goals	At the end of this course, the students will be able to: - Explain the basic principles of wind energy conversion. - Identify the steps required when developing offshore wind farms. - Differentiate between the challenges and opportunities of offshore and onshore wind energy harvesting. - Use existing tools for estimating the power produced by an offshore wind farm and the effect of wakes. - Integrate knowledge from various fields of engineering related to the development of offshore wind parks. - Reflect on the current and future trends in the field of offshore wind energy.	
Education Method	The teaching method uses the principle of flipped classroom. This means that, every week, the students have to perform online assignments (e.g. watch a lecture, read a text, etc) prior to coming to class. In class, additional lecture materials, exercises and feedback are given to the students.	
Assessment	The final assessment is a written exam. Additional points are given for performing the online activities prior to class.	

ET3034TU	Solar Energy	3
Responsible Instructor	Prof.dr.ir. A.H.M. Smets	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	In the course Solar Energy, you will learn to design a complete photovoltaic (PV) system for any application and location. This course introduces the technology that converts solar energy into electricity. The role of solar energy in both the energy transition towards a sustainable future and climate change mitigation will be discussed in detail. The physical principle of the PV energy conversion together with the design rules for a solar cell are introduced. These design rules are applied on the various PV technologies from cell up to module level. This program highlights various metrics related to performance, costs, reliability and circularity that will be used to evaluate the advantages and limitation of different PV technologies and its applications.	
Study Goals	After completion of the course, the students are ablo to: a) Reflect on the role of solar energy in both the energy transition towards a sustainable future and climate change mitigation b) Understand the physical working principles of photovoltaic energy conversion in solar cells c) Recognize and describe the various photovoltaic technologies and evaluate their applicability using various metrics related to performance, costs, reliability and circularity of PV. d) Design a complete photovoltaic system for any particular application and location	
Education Method	The education method is the so-called flipped-class room. Lectures will be provided in small video lectures on Brightspace. The on-campus lectures are focused on problem-solving. Students will actively work on problems that cover the entire content of the course and the various engineering skills. Solutions will be discussed during the lectures. Homework for every lecture includes: watching the relevant video lectures dedicated to the upcoming course and the remaining exercises of the previous lecture.	
Literature and Study Materials	Literature and Study Materials are provided on Brightspace: Videolectures Lectureslides. Large set of problems & exercises including answers. Answers are posted on Brightspace after the problems have been discussed in the lectures. Large set of examples of exams and answers. The book 'Solar Energy, basics, technology and systems' is free to download as e-book via all online retailers. Hardcopies of the book are sold at the ETV counter (building 36).	
Assessment	The course ET3034TU Solar Energy Energy will be assessed in Q2 and in Q3 (resit) of the academic year 2022/2023 as follows: The exam is a written exam. The exam takes 3 hours. The exam consist of open questions and multiple choice questions. A formula sheet will be provided during the exam (example is poste on Brightspace). You can use only pen, paper and a calculator. The exam will be assessed with a mark with a resolution of 0.5. To pass the exam a score equal to or higher than 58.0 out of 100.0 is needed.	
Remarks	Coordinator: Arno Smets	

WB3575	**Energy Conversion: Devices, Systems and Efficiencies**	4
Responsible Instructor	M. Ramdin	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic thermodynamics	
Course Contents	This course will introduce students to the thermodynamics of energy conversion processes such as steam power plants, gas/wind turbines, electrochemical/fuel cells, and heat pumps. Thermodynamics will be applied to evaluate the efficiency and performance of energy conversion processes. In addition, devices and components used for energy conversion and storage will be discussed. The course includes the following lecturing elements: A review of basic thermodynamics, including 1st and 2nd law, energy, entropy, and exergy balances, and properties of mixtures Entropy loss of real processes Exergy loss of real processes Energy Conversion CO2 Capture and conversion Solar Energy, Hydrogen and Fuel Cells Economic evaluation of energy conversion processes	
Study Goals	Study Goals At the end of the course, you should be able to describe the thermodynamics related to energy conversion. You will be able to apply entropy and exergy analysis to energy conversion systems, devices, and processes. Finally, you will be able to describe the operating principles of energy conversion processes and evaluate their economics.	
Education Method	(Online) lectures, video's, ungraded homework and (online) tutoring sessions, and graded group exercises	
Books	Moran & Shapiro: Principles of Engineering Thermodynamics, 9th Edition	
Assessment	The final grade will be based on the group exercises (30%) and written exam (70%). The minimum grade for the written exam should be at least 5.5. Participation in the group exercises is mandatory to pass the course.	
Department	3mE Department Process & Energy	

WB3580	Energy Storage	5
Responsible Instructor	Prof.dr.ir. W. de Jong	
Instructor	Dr. P.J. Vardon	
Instructor	Dr.ir. J.M. Bloemendal	
Instructor	Dr. R. Delfos	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Energy storage becomes increasingly important in society due to the introduction of increased shares of fluctuating renewable energy sources, like wind and solar energy. Moreover, a decreased dependency on fossil fuel resources is aimed at. This course will introduce the students to the different strategies and systems of energy storage (mechanical, thermal, chemical), with an emphasis on large-scale storage systems for electricity. The course includes the following lecturing elements: Introduction: context of renewable power supply (current and future) and scope. Mechanical storage systems (pumped hydro-electricity, compressed air energy storage, flywheels). Electrical storage systems (supercapacitors, superconducting magnetic Energy Storage). Electrochemical storage systems (batteries, flow batteries) Chemical storage systems (hydrogen production and storage options, power-to-gas, power-to-liquids, carbon sources). Carbon capture and storage/utilization. Thermal storage systems (sensible heat, phase change materials, thermo-chemical), including geothermal storage systems.	
Study Goals	The student can understand, select, and conceptually design and optimize processes for energy storage (both electricity and heat storage systems). More specifically, the student must be able to: 1. understand power supply and demand characteristics in relation to intermittent renewable energy sources and the role of energy storage herein; 2. understand and select appropriate systems for energy storage in terms of capacity and power and start-up and shut-down aspects; 3. solve problems related to mechanical, thermal, thermo-mechanical, electrical and chemical energy storage components of energy storage systems based on in depth knowledge of thermodynamics, mass and heat transfer, and (fluid) mechanics; 4. design basic process features for the abovementioned classes of energy storage systems.	
Education Method	Plenary lectures including cases and solving problems (4 hours per week) and separate working classes (2 hours per week); all foreseen to be on-line.	
Computer Use	the students have to use standard computer tools to calculate thermodynamic properties and performance of simple reaction and separation system	
Literature and Study Materials	Optional books: O.S. Burheim (2017) Engineering Energy Storage, Academic Press, 1st ed. G.Genta (1985), Kinetic Energy Storage, Butterworth R.A. Huggins (2010), Energy Storage, Springer Verlag A.F. Mills, (Basic) Heat (and Mass) Transfer, Pearson (all editions) or alike M.J. Moran, H.N. Shapiro, D.D. Boettner, and M.B. Bailey Principles of engineering thermodynamics (all editions) P.K. Nag (2014), Power Plant Engineering, McGraw-Hill, 4th ed. A.G. Ter-Gazarian (2010), Energy Storage for Power Systems, 2nd. ed. Towler and Sinnott, Chemical Engineering Design Principles, Practice and Economics, second edition. A. Vieira da Rosa (2013), Fundamentals of renewable energy processes, 3rd ed. Articles: Ausfelder et al. (2015) Energiespeicherung als Element einer sicheren Energieversorgung, Chemie Ingenieur Technik, 87(1-2), pp. 17-89. Lopes Ferreira et al. (2013) Characterisation of electrical energy storage technologies, Energy, 53, 288-298.	
Assessment	Written exam (on-line; individual), 50% of the final grade Assignment (group) on chemical energy storage, 25% of the final grade Assignment (group) on thermal energy storage systems, 25% final grade	
Permitted Materials during Tests	A standard (non programmable) calculator	
Percentage of Design	35%	
Department	3mE Department Process & Energy	

WB3585	Assessment of Energy Systems	3
Responsible Instructor	Dr.ir. L. Stougie	
Instructor	Dr.ir. G. Korevaar	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course introduces systematic ways to evaluate energy conversion and storage systems; life cycle assessment and exergy analysis are applied to cases.	
Study Goals	1) use thermodynamic knowledge, and especially the concept of exergy, in the field of large scale energy conversion and storage systems 2) perform a system assessment with the help of energy, exergy and LCA concepts and tools 3) evaluate large scale energy conversion and storage systems and make recommendations to improve these	
Education Method	Lectures and case studies	
Literature and Study Materials	Reader	
Assessment	The exam is on campus in the form of a closed book written exam. In case on campus education and examination is not possible, the exam will be in the form of a remote written exam with several fraud prevention measures.	
Permitted Materials during Tests	Simple, non-programmable calculator	
Tags	Energy Energy & Industry	

WB3595	Design Project Renewables Based Energy Conversion and Storage	12
Responsible Instructor	Dr.ir. J.T. Padding	
Contact Hours / Week x/x/x/x	x/x/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The finalisation of the ELECS minor consists of a project (12 ECTS per student) in which the students, in interdisciplinary group work, based on the boundary conditions of a renewable energy supply system, will conceptually develop, design and evaluate an energy conversion system with storage, or will develop and test a novel energy conversion/storage concept on a component level. This can be done via modelling and/or setting up and performing experiments. The design project will take place during the 1st and 2nd period (Q1 and Q2) of the academic year. During Q1 (3ECTS) the students will more precisely define the goal of the design, do a literature search, and do preparatory work (including ordering materials, parts and sensors in case of experimental setups). Q1 will be closed with an oral presentation of the design plan. The actual design and evaluation of the process or component will take place during Q2 (9 ECTS). Q2 will be closed with a final report, a working model or prototype, and a pitch and poster presentation at the ELECS final symposium. NOTE: Depending on the COVID-19 situation, some or most meetings in the 2022-2023 edition may be planned to occur online. We will keep you updated through Brightspace.	
Study Goals	After completion of the course, the students are able to: - apply and extend the fundamental knowledge that plays an important role in sustainable energy conversion and storage (in particular of thermodynamics, heat transfer, reaction kinetics and fluid mechanics) in designing a process. - divide a given process in sub-processes, together forming the entire process - apply thermodynamics to derive a process design of a technical system and to optimize the thermodynamic design (not for all projects). - apply heat transfer concepts to determine the main dimensions of heat exchangers (not for all projects) - apply reaction kinetics concepts to derive the main dimensions of reactors (not for all projects) - develop a list of requirements for the process design, accounting for its sustainability and limits to ethical, societal and safety - evaluate a design according to sustainability criteria which have to be formulated - write a well-structured technical report - in case of an experimental design project: deliver a working prototype - in case of a simulation design project: deliver a working model in Matlab, Python, or a programming language like C or Java - explain and defend their design in an oral (poster) presentation	
Education Method	During 14 weeks, students work in small groups of 4 to 5 students on a design project. The projects will be supervised by permanent staff of the research groups contributing to the minor. Dedicated instructions will be given on aspects of modelling and simulation and/or laboratory practices (a.o. safety). Note: depending on the COVID-19 situation, in the 2022-2023 edition meetings may be planned to occur online. Experimental setups will be constructed by at most 2 students at a time, where the other students in the team may focus on modelling the same system and data analysis.	
Assessment	In Q1, week 4, the students need to deliver a 2-to-3-page summary of the literature search. This will be evaluated by the supervisors (weight 10%). In Q1, week 9, the students need to deliver a 8-minute design plan presentation (i.e. the motivation, precise goal, results of the literature search, and an outline of their approach), followed by 3 minutes of questions. The presentations of all student groups will be organised in a joint session with as many supervisors present as possible. The presentation will be evaluated by the supervisors (weight 15%). In Q2, week 9, the students need to deliver a final report and a working model or prototype. This will be evaluated by the supervisors (report has weight 40%, model or prototype has weight 20%). A few days later in Q2, week 9, during the ELECS final symposium, the students will give a short pitch of their design in 4 minutes, followed by presentation of more details in a poster session. The pitch and poster presentation will be evaluated by the supervisors (weight 15%).	
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